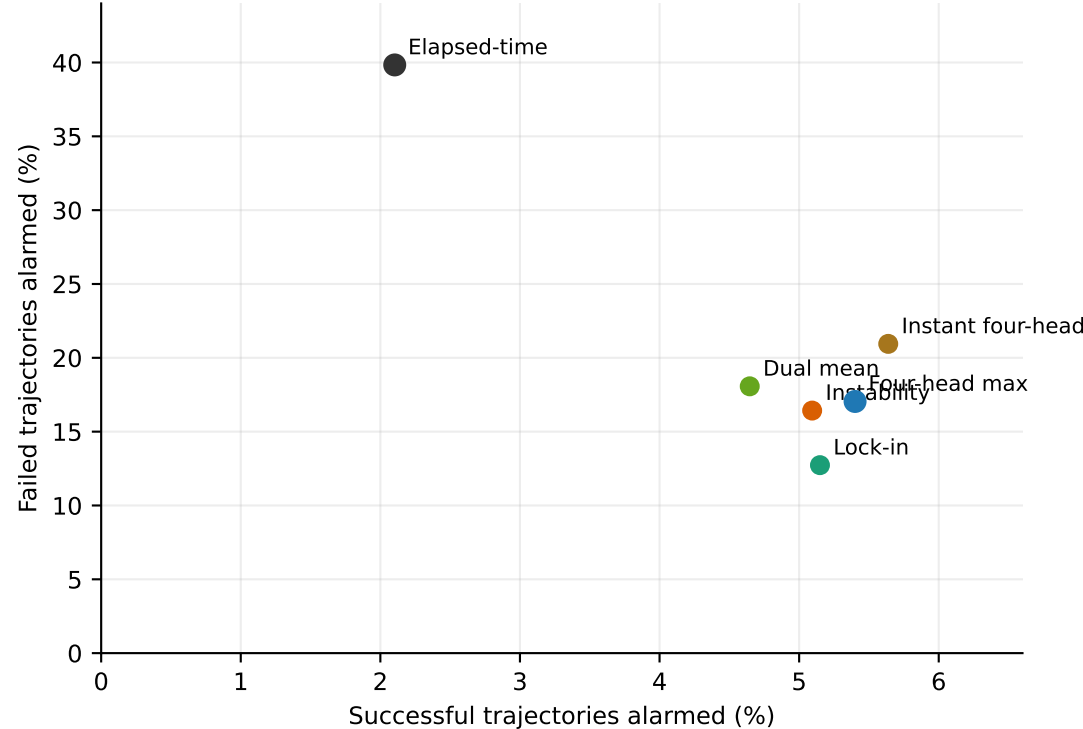
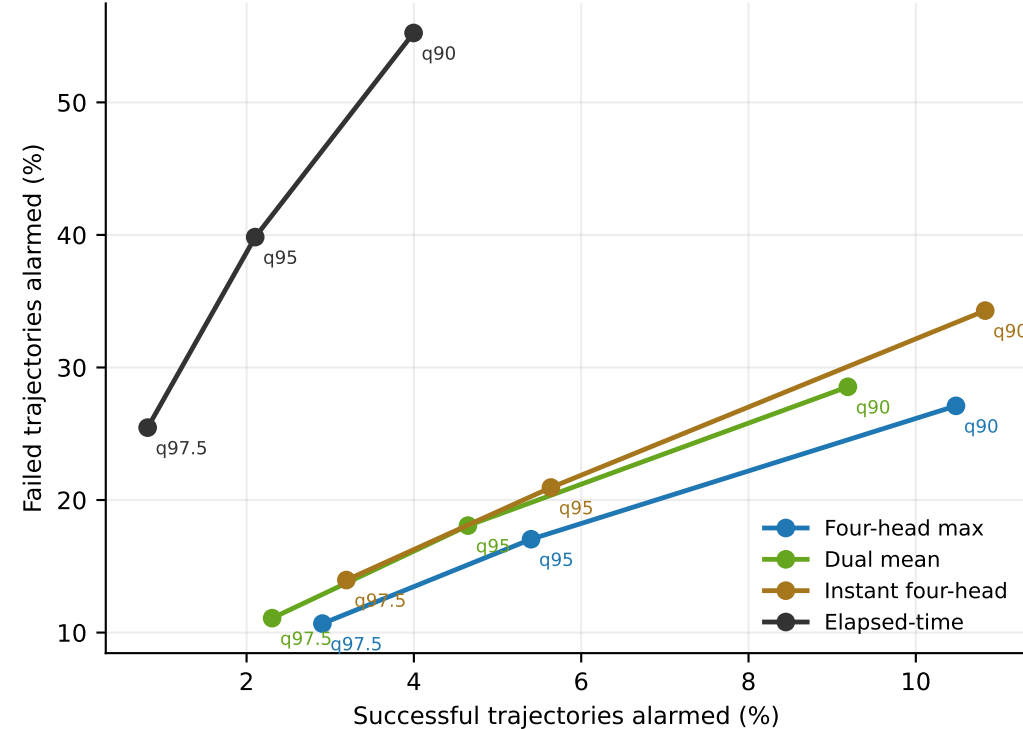


Train-free causal MoE alarm: phenotype information, weak overall trigger
Main: 14,800 trajectories, 487 failures. External: 1,200 trajectories, 45 failures; four-head recall 60.0%, FPR 3.7%.

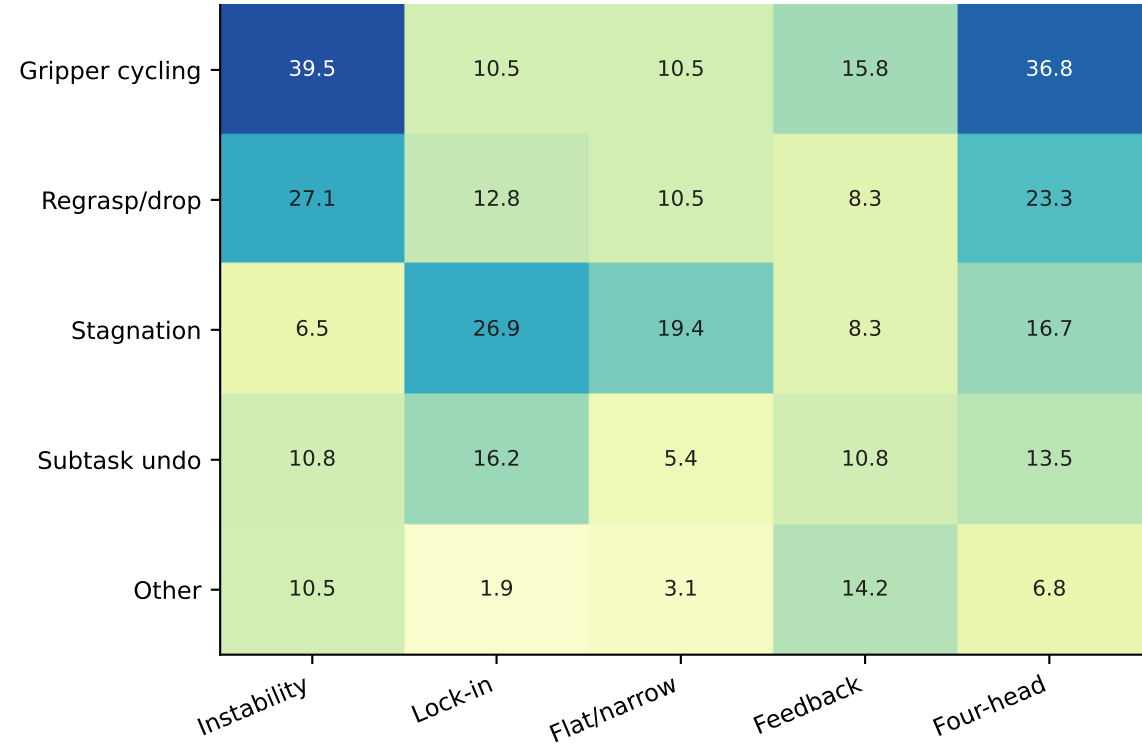
A Main cohort at unlabeled q95



B Frozen alarm-budget sweep



C Failure subtype alarm rate at q95 (%)



D Four-head timing around physical proxy onset

